Yenepoya Institute Of Arts Science Commerce and MAnagenent

Project – Low Level Design (LLD)

PG Recommendation System for Smart Cities

Student Name: Shaima Sharif Kulavoor
23BSAMD030
LH - 08

BSc Artificial Intelligence,
Machine Learning, Data Science
(IBM, TCS)

Industry Mentor: Mr. Sumit Kumar Shukla
Guided by: Ms. Ankitha Poojary

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1. Introduction

1.1 Scope of the document

This document describes the low-level design of the PG Recommendation System. It explains internal modules, logic, data handling, and implementation details.

1.2 Intended Audience

- Developers
- Project evaluators
- Faculty reviewers

1.3 System Overview

The system recommends PG accommodations based on:

- City
- Budget
- Tenant type

It uses a dataset and filtering logic to generate recommendations.

2. System Design

2.1 Application Design

The system is implemented using:

- Python
- Pandas
- Jupyter Notebook

Architecture Type:
Single-user, rule-based system

2.2 Process Flow

User inputs:

- City
- Maximum Rent
- Tenant Type

System loads dataset (CSV)
Data is filtered based on:

- Matching city
- Rent <= budget
- Tenant preference

Filtered results are displayed

2.3 Information Flow

User Input → Processing Module → Filtering Logic → Recommendation→Output

2.4 Component Design

1. Input Module

Takes user input using input()
Stores values in variables

2. **Data Loading Module**

Loads dataset using:

```
import pandas as pd
df = pd.read_csv("House_Rent_Dataset.csv")
df.head()
```

3. **Data Processing Module**

```
# Filter dataset based on user input
filtered = df[
    (df["City"].str.contains(city, case=False)) &
    (df["Rent"] <= budget) &
    (df["Tenant Preferred"].str.contains(tenant, case=False))
]
```

4. **Recommendation Engine**

- Applies rule-based filtering
- Combines multiple conditions
- Returns matching rows

5. **Output Module**

Displays formatted PG recommendations

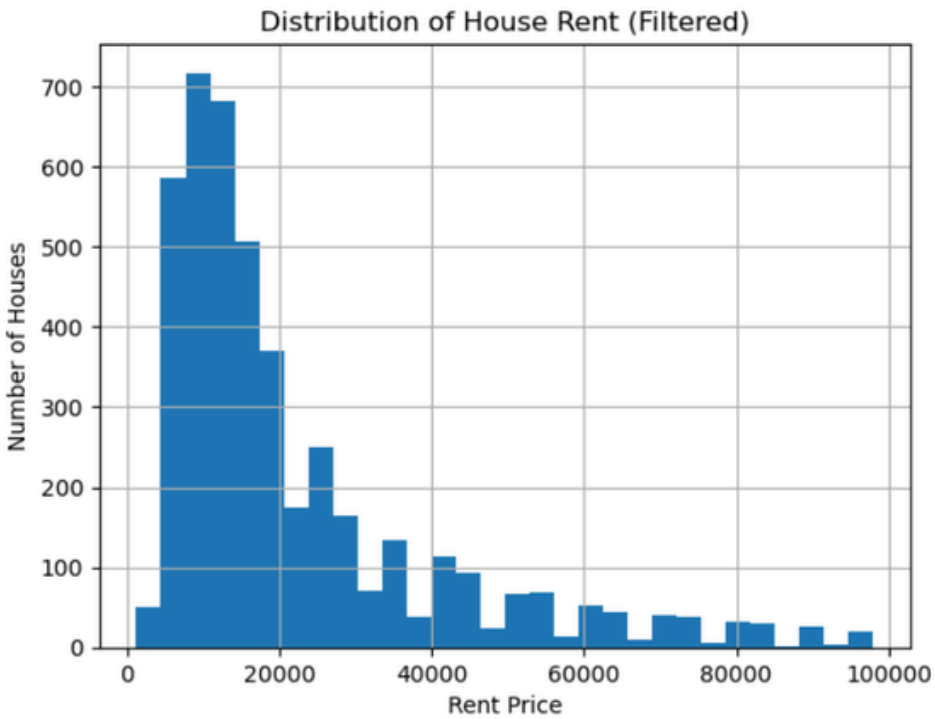
Shows:

- Location
- Rent
- BHK
- Furnishing
- Tenant type

```
pg_agent()
```

```
Smart PG Recommendation Agent
--------------------------------
Enter city:  Bangalore
Enter maximum rent:  11000
Tenant type (Bachelors/Family):  Bachelors

Recommended PG Accommodations:

Nagappa Reddy layout-Dooravani Nagar | Bangalore | BHK: 2 | Rent: ₹10500 | Furnishing: Semi-Furni
shed | Tenant: Bachelors/Family
Talaghattapura | Bangalore | BHK: 2 | Rent: ₹10000 | Furnishing: Furnished | Tenant: Bachelors/Fa
mily
Rajajinagar | Bangalore | BHK: 1 | Rent: ₹7500 | Furnishing: Unfurnished | Tenant: Bachelors/Fami
ly
Ramamurthy Nagar | Bangalore | BHK: 2 | Rent: ₹10500 | Furnishing: Semi-Furnished | Tenant: Bache
lors/Family
Thanisandra | Bangalore | BHK: 1 | Rent: ₹8000 | Furnishing: Semi-Furnished | Tenant: Bachelors/F
amily
```

```
df[df["Rent"] < 100000]["Rent"].hist(bins=30)

plt.title("Distribution of House Rent (Filtered)")
plt.xlabel("Rent Price")
plt.ylabel("Number of Houses")

plt.show()
```



2.5 Key Design Considerations

- Simplicity and readability
- Efficient filtering using pandas
- Scalable for ML integration in future

2.6 API Catalogue

(Not applicable – system is standalone for now

3. Data Design

3.1 Data Model

Dataset Fields:

- City
- Rent
- BHK
- Furnishing Status
- Tenant Preferred
- Location

3.2 Data Access Mechanism

- Data is stored in CSV file
- Accessed using pandas

3.3 Data Retention Policies

- Static dataset
- No user data stored

3.4 Data Migration

Not required (initial version)

4. Interfaces

- Command-line interface (CLI)
- User interacts via terminal

5. State and Session Management

- No session handling
- Stateless system

6. Caching

Not implemented

7. Non-Functional Requirements

7.1 Security Aspects

No sensitive data used

7.2 Performance Aspects

- Fast filtering using pandas
- Suitable for small to medium datasets

8. References

- Python Documentation
- Pandas Documentation